# Who is who: what matters in injuries in professional team sports

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**Abstract**

**Objective** Per-hour injury rates guide rest and match calendars. We measured how injury rates differ between players and with recent load, and who plays the extra minutes after a heavy week.
**Methods** Observational study of public match records from nine team sports, both sexes. Injury records covered 22 573 players (748 women) in six sports. We compared differences between players with changes within the same player. A heavy week had a full match or more within seven days; a longer-gap week followed an 8-to-13-day gap.
**Results** Between two players picked at random, the higher usual injury rate was typically 1.82 times the lower. After a full match, the same player's rate typically differed by only 5%. As expected, two or more injuries of 28 days or more in three years went with 1.3 to 1.7 times the rate. Players who played more in the previous week are known to play longer. But within the same player, playing time rose less. Starters with more recent load stayed on the field longer. These patterns held in every sport we could test, whether or not a replaced player may return.
**Conclusion** Injury rates differ far more between players than with recent league load. Heavy weeks fall mostly to fit regulars, so per-hour rates that compare players mostly show who carries the load. Compare each player's heavy weeks with the same player's longer-gap weeks, season by season. Medical staff can read repeated injuries of 28 days or more as a risk marker.

## What is already known on this topic

- Players differ in injury risk beyond their load and their previous injuries.
- Per-hour injury rates assume that the hours a player records do not depend on the load under study.

## What this study adds

- Per-hour injury rates that compare players mostly show who carries the load: in every sport we could test, rates differed far more between players than with recent load, and after a heavy week, playing time rose less within the same player than across players.

## How this study might affect research, practice or policy

- Researchers and injury surveillance programmes should compare each player's heavy weeks with the same player's longer-gap weeks in the same season and, where they infer injury from team sheets, should not read a missed start in the next match as injury.
- Club medical staff can read two or more injuries of 28 days or more in three years as a marker of higher risk, which confirms a known risk factor: players with this history came through most matches uninjured, but were injured more often.

## Introduction

Injury rates guide which players medical teams rest and how many matches players play.[1, 2] Both uses assume that recent load, meaning last week's play, drives injury risk. Yet some players are injured more often than others. A previous injury, for example, is a strong risk factor for the next one.[3] Suppose the players who play the most are also those injured least. Then injury rates that compare players will make a heavy week, a full match or more in the previous week, look safe.

In football, injuries are counted per 1000 hours of play.[4–6] Studies of fixture congestion and recent load use such rates.[7–12] All of them assume that the hours a player records stay the same, whatever the recent load. But players who played more in the previous week play more minutes in the next match.[13, 14] We call this rise the workload gradient.

A per-hour rate divides injuries by the minutes played. The number we divide by is called the denominator. After a heavy week the minutes rise, so the denominator rises too. The rise could

have two causes. A heavy week might change how long a player plays. Or the players with a heavy week might be the regular starters, those the coach picks to start every week and keeps on the field. We call this choice team selection. Which cause holds decides whether a per-hour rate compares loads or compares players.

Earlier work shows that players differ in injury risk beyond their load and their previous injuries.[15] In basketball, coaches rest players when their injury risk rises.[16] In medicine, comparing groups of people who were picked in some way can mislead.[17] But to our knowledge, only the first author's nine-sport preprint[18] has measured how much of the recorded playing time comes from team selection. Others have raised the question.[1, 19]

In nine team sports, we asked how much injury rates differ between players and with recent load. We also asked why comparing players can mislead.

# Methods

## Design and data

This observational study used public match records from nine team sports: it studied matches as they were played. The records show each team's line-up and the minute of each substitution. The study follows the STROBE reporting guideline.[20] Table 1 lists the sports and names the leagues (Supplementary Section S2). The records of men's football come from Transfermarkt, which we call the provider. We asked the same questions of 37 more leagues in seven sports (Supplementary Section S9). A league sample is one league's seasons under one substitution limit, since most football leagues moved from three substitutions per match to five in 2020.[21]

## Recent match load

Recent match load, or recent load for short, is the league minutes a player played in the previous seven days. It is counted in full matches of 90 minutes. A player with a full match or more was loaded: a heavy week. A player with no minutes was rested. A longer-gap week is a rested week in which the club had no league match, 8 to 13 days after its previous one. Then the fixture list set the rest, rather than the coach or an injury. So a longer-gap week makes a fair comparison for a heavy week (Figure 1C).

## Injury records

A substitution is injury-coded when the match report gives injury as its reason. We used only league-seasons that allowed five substitutions, and in which at least 90 per cent of substitutions have a reason. A recorded absence after an injury-coded substitution is a sign of real injury. So we also counted only the injury-coded substitutions that a recorded absence followed. We call this second count the narrower outcome. Every football injury result is given on both outcomes (Supplementary Section S3). In the other sports, an injury is each league's own record of an absence (Supplementary Section S12).

A statistical model gave each player their own usual injury rate, which we call the baseline rate. It takes in everything that stays the same about a player, such as position, usual load and how the club records injuries. The same model measured how a player's own rate differed with each extra full match of recent load. We ran it in each of 11 player groups (Table 1; Supplementary Section S11). In five sports, the records also give each player's injury history (Supplementary Section S12).

## The injury comparison

We counted injury-coded substitutions per 1000 hours played as a starter, which we call starter hours. We compared heavy weeks with longer-gap weeks in two ways. Across starters, we compared different starters' weeks in each league-season, and then combined the league-seasons (Mantel–Haenszel). Within each starter, we compared each starter's heavy weeks with the same starter's

own longer-gap weeks in the same league-season (conditional Poisson regression). Figure 2 adds other counts, other comparison weeks and checks (Supplementary Section S6).

### The workload gradient

The workload gradient is how much a player's recorded minutes rise with recent load. We split it into two parts: one comes from who starts, the other from starters and substitutes playing longer.[22] For starters, the within-player slope is the gradient measured within each starter. It shows whether the same starter played longer after their own heavy weeks (Supplementary Section S3).

### When starters came off

We modelled the chance that a starter was taken off, by their recent load (Supplementary Section S3). We ran this model in the eight European men's leagues, in a worldwide survey of 162 samples from 99 first-tier men's competitions, and in the other leagues that recorded substitution times (Table 1; Supplementary Sections S8 and S9). In the eight leagues, a second version compared each starter's heavy weeks with the same starter's other weeks.

### Statistical analysis and pre-set criteria

The statistics follow the CHAMP statement.[23] We set two criteria before testing them (Supplementary Section S4). First, from sample to sample, the within-player slope would vary at most half as much as the rise across starters. Second, the more firmly a league keeps loaded starters on, the more of the starters' rise would come from differences between starters. Every other analysis, the injury comparison among them, is exploratory: it finds leads to check. But we chose the comparison the design favours, each starter against themselves in the same season, after seeing the results of similar comparisons. Each result is reported on its own, as if it were the only comparison. We wrote the analysis scripts with an AI assistant and checked them against hand-worked tests.

### How to read the results

Most results are ratios of two injury rates. A ratio of 1.16 means a rate 16% higher than the rate it is compared with, and 0.88 means 12% lower. Each ratio comes with a range, its 95% confidence interval (95% CI). The true value is likely to be in this range. When the whole range is above one, or below one, the difference is clear. When the range includes one, there is no clear difference. For heavy weeks, we call a clear rise riskier and a clear fall safer.

### Equity, diversity, and inclusion statement

The league samples include women's football leagues, and women's basketball and cricket. The worldwide survey covers men's leagues only. Women's injury records come from one league, the WNBA: 748 of the 22 573 players with injury records. The two authors work at one European university, in statistics and medicine.

### Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

## Results

### Injury rates differed far more between players than with recent load

Between two players picked at random from one season, the higher baseline rate was typically 1.82 times the lower (Figure 1A). After a full match, the same player's rate typically differed by only 5%,

Table 1: Which data answer which question. The nine sports are football, rugby union, Australian rules, basketball, ice hockey, American football, rugby league, baseball and cricket. The eight European men's leagues are those of England, France, Germany, Italy, the Netherlands, Portugal, Spain and Turkey. League-seasons, players and sizes are totals over the samples. In the first rows, the league samples are the 11 player groups. A player in two samples of one question counts twice, unless the source gives each player one identifier: the NBA with the G League, and MLB's batters with its pitchers. The worldwide survey, the injury comparison and the next start count each player once. The survey's pooled results combine the 140 samples played with three or five substitutions allowed (Supplementary Table S24). The next start asks how often a starter taken off early started the club's next match (Supplementary Section S3). Outside football, an injury is each league's own record, counted once per absence (Supplementary Section S12). Median age is the range of the samples' median ages at appearance; a dash marks rows without one (Supplementary Table S2). The injury comparison's row includes goalkeepers, whom Figure 1C and Figure 2 leave out (19 231 starts).

| Question | Sport and sex | League samples | League-seasons | Players | Of them, women | Median age, years | Size | Unit of size |
|---|---|---|---|---|---|---|---|---|
| Players against recent load (Figure 1A) and injury history (Figure 1B) | Football, men: England, France, Germany, Italy and Spain, as one group | 1 | 27 | 4591 | 0 | – | 6710 | injury-coded substitutions, outfield starters |
| | Australian rules, men | 1 | 9 | 1253 | 0 | – | 1824 | absences through injury |
| | Basketball, men and women | 4 | 62 | 4252 | 748 | – | 3180 | absences through injury; injured-list placements in Japan |
| | Ice hockey, men | 2 | 13 | 1932 | 0 | – | 1946 | absences through injury |
| | Baseball, men, batters and pitchers | 2 | 32 | 3965 | 0 | – | 4743 | injured-list placements |
| | American football, men | 1 | 13 | 6580 | 0 | – | 4613 | absences through injury |
| Who starts (Figure S3A) | Football, 10 men's and 7 women's; rugby union, men | 18 | 74 | 18 419 | 2104 | 24.5–28.7 | 685 148 | appearances |
| Within the same player (Figure S3B) | Football, men and women; rugby union, men; Australian rules, women | 25 | 57 | 18 673 | 3265 | 24.4–28.5 | 14 963 | players with two or more starts |
| Kept on, worldwide survey | Football, men's first tiers in 96 of 99 competitions, the eight European men's leagues fitted apart | 140 of 162 | 538 | 72 208 | 0 | – | 2 089 384 | starter appearances |
| The injury comparison (Figures 1C and 2) and one start in 31 | Football, men, the same five leagues | 5 | 27 | 5942 | 0 | 25.4–27.4 | 296 029 | appearances, goalkeepers included, with 7158 injury-coded substitutions |
| The next start | Football, men, 14 leagues, five substitutions | 14 | 82 | 13 202 | 0 | 24.8–28.7 | 490 125 | starts followed by another club match |

up or down. This held in every sport with injury records (Supplementary Section S11).

As expected, injury history also marked higher risk. Players with two or more injuries of 28 days or more in the previous three years had 1.3 to 1.7 times the injury rate of the others. This held in every sport with injury histories (Figure 1B; Supplementary Section S12).

### Heavy weeks looked safer across starting players than within each one

In European men's football, the standard comparison made heavy weeks look safe. It compares different starters with each other, per hour on the field. Across starters, heavy weeks looked 12% safer than longer-gap weeks (0.88, 95% CI 0.82 to 0.94; Figure 2). Injuries confirmed by a recorded absence showed no clear difference (0.94, 0.85 to 1.05).

Within the same starter and season, the picture turned. Heavy weeks had 6% more injury-coded substitutions, with no clear difference (1.06, 0.99 to 1.13). Injuries confirmed by a recorded absence were a clear 16% more frequent (1.16, 1.03 to 1.29).

### Who starts accounts for most of the workload gradient

The extra playing time after a heavy week came mostly from who starts. Who starts accounted for 73% to 92% of the workload gradient in all 18 samples of football and rugby union (Supplementary Figure S3A). Within each starter, recorded minutes changed little (Supplementary Figure S3B). In the 37 other leagues, playing time also rose less within the same player than across players, in all 54 player groups (Supplementary Section S9). They include basketball and ice hockey, where players may return, so a player's minutes are free to rise.

### Across starters, those with a full match in the previous week stayed on longer

Coaches also kept loaded starters on the field. Starters with a full match in the previous week were taken off less often. This was clear in all 162 samples of the worldwide survey, and in all 15 other leagues that recorded substitution times (Supplementary Table S26). In the eight European men's leagues, it was also clear within the same starter (Supplementary Table S17). Part of this result is about roles. Some starters play the whole match every week (Supplementary Section S6).

A summary of our key findings is presented in Figure 3.

## Discussion

Injury rates differed far more between players than with recent load. Heavy weeks fall mostly to regular starters, the players fit enough to keep their place. So in European men's football, heavy weeks looked safer across starters than within each one.

### Per-hour rates divide by who starts

In football and rugby union, most of the extra minutes after a heavy week came from who starts. So a per-hour rate compares different players, as well as different loads.[24] For a regular starter, a heavy week is the usual week, so most heavy weeks belong to regular starters. Workplace health research calls this the healthy worker effect: the workers who do most of a job are those fit enough to stay in it.[25–27]

Other counts have been proposed. Some injury research has corrected its rates by counting only the players fit to play.[28] Rates per appearance and per hour can also give different answers.[29–31] Two recent papers in this journal discuss counting per appearance.[32, 33] That count does not remove the selection among starters (Figure 2). Measuring how much minutes rise with load, as our earlier preprint did,[13] cannot show who carries the load either.

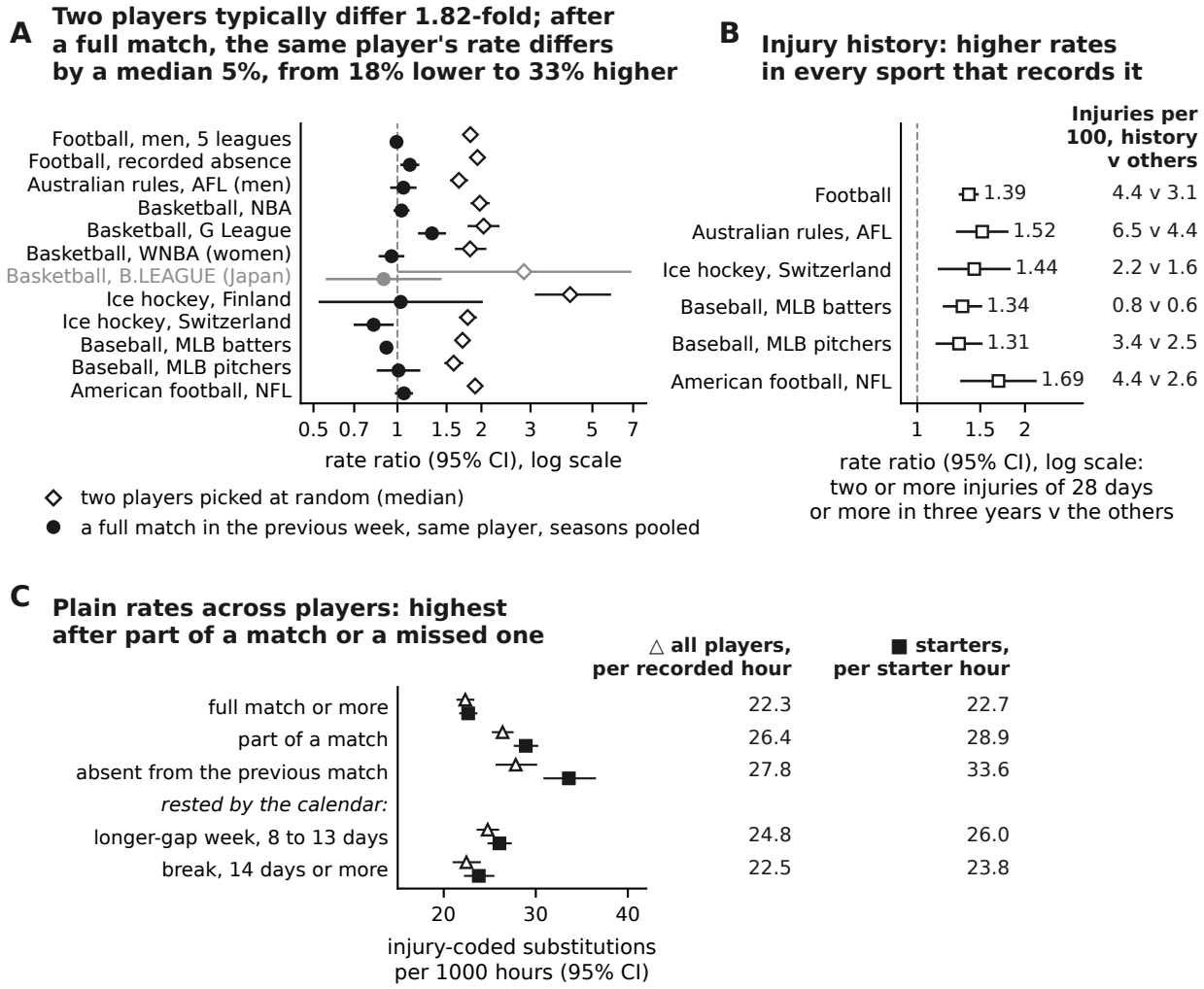



Figure 1: Injury rates differed far more between players than with recent load. (A) For each player group, the open diamond shows how far apart two players picked at random typically are: the median rate ratio, the higher rate over the lower. The filled dot shows how the same player's rate differed after a full match in the previous week, with seasons pooled. Lines show 95% CIs (Supplementary Table S29). On the log scale, doubling and halving sit the same distance from one. Football counts outfield starts only, once for injury-coded substitutions and once for those a recorded absence followed. The G League leaves out absences that were likely moves to the NBA. Japan's B.LEAGUE, drawn lighter, had only 27 injuries, too few to tell its two measures apart. Finland's ice hockey league, the Liiga, has the highest ratio between players, because its teams record injuries in different ways (Supplementary Section S12). Outside football, some injury lists record rest, or a move to another team, as an injury, which would move both markers to the right. (B) Players with two or more injuries of 28 days or more in the previous three years, against the others: the rate ratio, and beside it each group's injuries per 100 starts or appearances (Supplementary Table S30). (C) Plain injury rates, before any comparison within players: injury-coded substitutions per 1000 hours among outfield players in the 27 league-seasons of the injury comparison, with 95% CIs. Triangles count all players per hour played, and squares count starters per starter hour. Part of a match is some minutes, but less than a full match. A player was rested by the calendar when the club had no league match that week, and absent when they missed the club's previous match.

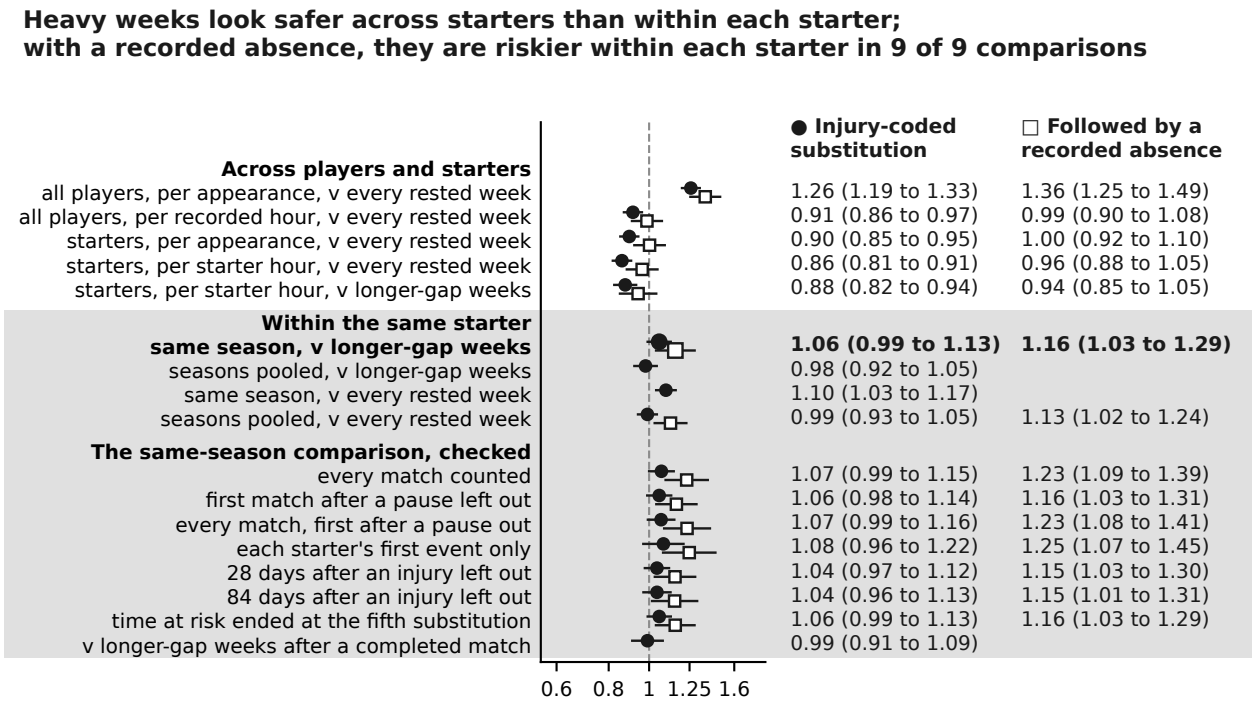



Figure 2: Heavy weeks looked safer across starting players than within each one. Outfield players in the 27 league-seasons of the injury comparison. Each row gives the ratio after a full match or more in the previous week, with 95% CIs. Filled circles count injury-coded substitutions, and open squares count those a recorded absence followed. Heavy weeks look safer where the whole interval lies below one, and riskier where it lies above one. Unshaded rows compare all players, or all starters, with each other (Mantel–Haenszel), per appearance, per hour played or per starter hour, against every rested week or against longer-gap weeks. Shaded rows compare each starter's own weeks (conditional Poisson regression), within the same league-season or with seasons pooled. The row in bold is the comparison the design favours, because it sets each starter against themselves in the same season. The rows under it repeat it with other rules (Supplementary Section S6). Three rows have no square: those comparisons were run on injury-coded substitutions only. Supplementary Table S20 gives the counts.

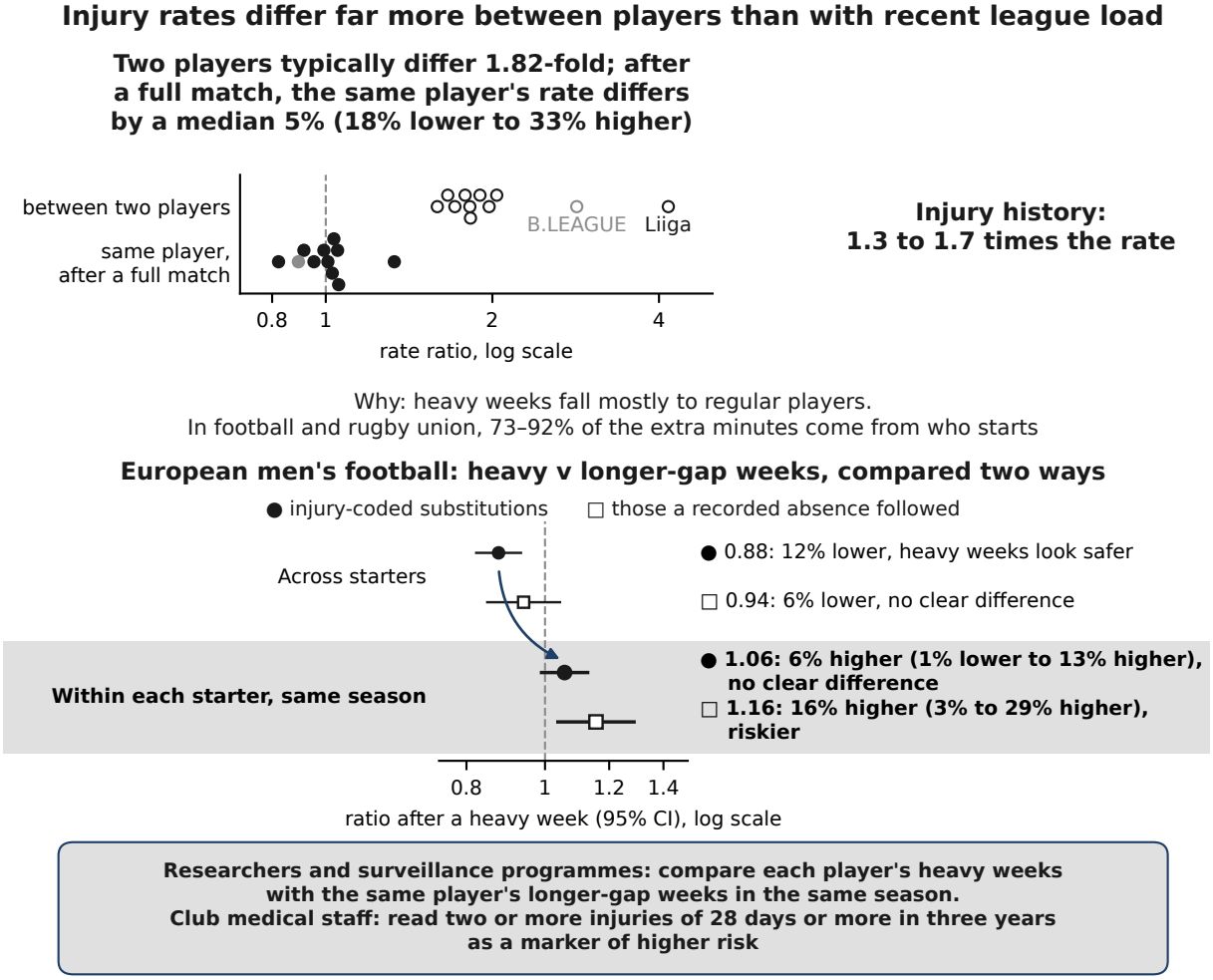



Figure 3: Central Figure: injury rates differ far more between players than with recent league load, and how to compare like with like. In the top row, each dot is one of the 11 player groups. An open dot gives the median rate ratio between two players picked at random. A filled dot gives the same player's rate ratio after a full match in the previous week, seasons pooled (Supplementary Table S29). The open dot at the far right is Finland's Liiga, whose teams record injuries in different ways. Japan's B.LEAGUE, drawn lighter, had too few injuries for its two measures to be told apart. The injury history's ratios span the six groups that record it. In the second row the symbols change meaning. A filled mark shows outfield starters' injury-coded substitutions per starter hour, and an open mark shows those a recorded absence followed. Both are against longer-gap weeks: across starters, and within each starter (shaded) in the same league-season. The words beside each interval give its reading: safer or riskier when it lies wholly below or above one, no clear difference when it holds one. The arrow runs from the standard comparison, across starters, to the comparison within each starter. A heavy week had a full league match or more in the previous seven days. A longer-gap week followed a gap of 8 to 13 days. The box gives this study's own advice.

### How comparing players misleads

- **Heavy weeks look safe.** Comparing starters with each other gives the heavy week the credit for the regular starters' lower risk. This matches a healthy worker effect.[34]
- **The denominator can flip the answer.** Loaded players had 26% more injury-coded substitutions per appearance than rested players, but 9% fewer per hour played (Figure 2).
- **Comparing players by their load misses those at higher risk,** such as players with repeated injuries of 28 days or more.

### Other factors in resting a player

Besides recent load, a decision to rest a player can weigh:

- previous injury, a strong risk factor;[3]
- recovery time: among rested starters, rates were lowest after breaks of 14 days or more;
- every match played, and the travel between them;[10]
- training load, which match records never show;[11]
- squad depth and rotation.[8, 35]

### Implications for research

First, the comparison the design favours sets each player against themselves in the same season. Report it beside the standard comparison across players. It removes stable differences between players, but selection that changes from week to week remains.

Second, count injuries confirmed by a recorded absence, and count load over every match played (Figure 2).

Third, researchers who infer injury from team sheets should not read a missed start as injury (Supplementary Section S3).[36]

### Clinical implications

First, two or more injuries of 28 days or more in three years mark a group at higher risk, not the players who will be injured.[37] The history confirms a known risk factor and should not by itself change selection (Supplementary Section S12).

Second, a small difference with recent load does not show that congestion is safe. Within the same starter, the narrower outcome was 16% more frequent after a full match. This result comes from one sport and from an exploratory comparison.

### Limitations

The most important limitation is how recent match load was measured.[38] The main analysis counts league matches only. Matches in the international breaks are never counted. So an international player's load is understated in exactly the weeks that congestion research studies (Supplementary Table S12).

Second, the records are public match data, not surveillance data kept by club doctors.[39, 40] The injury reason is a label from the match report, not a diagnosis. It also covers players taken off as a precaution. A recorded absence followed injury-coded substitutions a little more often after a heavy week. So the narrower outcome may make heavy weeks look slightly riskier (Supplementary Section S3). No video audit of the study seasons was feasible. The provider also lists every substitution made in added time at minute 90. So recorded minutes leave out added time, and a starter taken

off then seems to have played the whole match (Supplementary Section S3). Giving the added time back changed the per-hour ratios little (Supplementary Section S6).

Third, the design is observational: it records what happened and changes nothing.[41] Selection from week to week can still bias the result. For example, a player who came off early with a knock may carry more risk into the next start. A computer simulation measured that bias, and the real data showed no sign of it (Supplementary Section S6). In football and rugby union, recorded minutes stop at the full match, which sets the smallest within-player slope the design could detect (Supplementary Section S6).

Fourth, we saved the two pre-set criteria, with a date, to the project's own files, not to a public protocol. A public protocol is a plan published in advance. We saved both after our earlier preprint had used the same records. The first passed narrowly and the second failed. On the first, the within-player slope varied 0.48 times as much as the rise across starters, against a limit of one half. The second failed because its link was too uncertain to tell either way (Supplementary Section S4). We also registered a confirmation on the next complete seasons, but the study ends before those seasons do.

Finally, the samples differ in sport, sex, era and size. The only women's injury record is one basketball league. The injury comparison comes from European men's football (Supplementary Section S9).

## Conclusion

Injury rates differ far more between players than with recent league load. Heavy weeks fall mostly to regular players, who are fit enough to keep their place. So per-hour injury rates that compare players mostly show who carries the load. Within each starter, injury-coded substitutions confirmed by a recorded absence were more frequent after a full match. This needs testing on new seasons. Compare each player's heavy weeks with the same player's longer-gap weeks in the same season. Read repeated injuries of 28 days or more as a marker of risk.

## Declarations

**Contributors** GPR designed the study, collected and analysed the data and wrote the manuscript. NM shaped the methods and the clinical framing of the earlier preprint that this manuscript extends. Both authors approved the final version. GPR is the guarantor. Anthropic's Claude, a generative AI assistant, was used under GPR's direction to write and check analysis code, search the literature and edit the text. It was used because the first author could not otherwise build and check analyses of this size. The study design, the claims and their interpretation are the authors'. Every reported quantity is computed from the committed data by the archived code and was verified by GPR. Both authors take full responsibility for the content.

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**Competing interests** None declared.

**Prior dissemination** This manuscript is posted on arXiv as the next version of our earlier preprint.[13] That preprint raised the question this manuscript answers. Some of this manuscript's results were first posted in the first author's nine-sport preprint.[18]

**Ethics approval** The study reused only public, pre-existing match records. There was no recruitment, no contact with any individual, no intervention and no access to medical records or clinical files, so informed consent did not apply. The records name professional athletes, and some reasons for substitution are injuries. They were therefore treated as special-category personal data despite being public. Processing was for scientific research under the research provisions of the General Data Protection Regulation. Everything published, every table, figure and deposited file, is aggregated to whole league samples, so none of it describes a single player. Ethical approval was not sought.

These published records were treated as archival data, which Trinity College Dublin's research ethics guidance excepts from approval.[42] The same guidance grants no approval retrospectively.
**Data availability statement** Data are available in a public, open access repository. The aggregated results behind every table and figure, and the analysis scripts, are in the archived record at DOI RESERVE-BEFORE-SUBMISSION. No player-level record is shared. The match records the scripts read identify players, so they are not redistributed. Their reconstruction from the public providers, each source's terms of use as read and any permission held are documented in the archive. Statistics from NBA.com are credited to NBA.com, as its terms ask.

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